Question 1:

Solve the differential equation $y' + 5xy = 2x$

Note: The differential equation is of the form $\dfrac{dy}{dx} + M(x)\cdot y = N(x)$.

a) $e^{\int M(x)\cdot dx} = \underline{\quad ? \quad}$

b) $\int N(x)\cdot e^{\int M(x)\cdot dx}\cdot dx = \underline{\quad ? \quad}$

c) $y = \dfrac{\int N(x)\cdot e^{\int M(x)\cdot dx}\cdot dx \; + C}{e^{\int M(x)\cdot dx}} = \underline{\quad ? \quad}$

Question 2:

Solve the differential equation $y' - 3xy = 4x$

Note: The differential equation is of the form $\dfrac{dy}{dx} + M(x) \cdot y = N(x)$.

a) $e^{\int M(x) \cdot dx} = \underline{\quad ? \quad}$

b) $\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx = \underline{\quad ? \quad}$

c) $y = \dfrac{\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx \ + C}{e^{\int M(x) \cdot dx}} = \underline{\quad ? \quad}$

Question 3:

Solve the differential equation $y' - 6x^2 y = 4x^2$

Note: The differential equation is of the form $\frac{dy}{dx} + M(x) \cdot y = N(x)$.

a) $e^{\int M(x) \cdot dx} =$ ___?___

b) $\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx =$ ___?___

c) $y = \dfrac{\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx \ + C}{e^{\int M(x) \cdot dx}} =$ ___?___

Question 4:

Solve the differential equation $y' + 12x^3 y = 2x^3$

Note: The differential equation is of the form $\frac{dy}{dx} + M(x) \cdot y = N(x)$.

a) $e^{\int M(x) \cdot dx} = \underline{\quad ? \quad}$

b) $\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx = \underline{\quad ? \quad}$

c) $y = \dfrac{\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx \ + C}{e^{\int M(x) \cdot dx}} = \underline{\quad ? \quad}$

Question 5:

Solve the differential equation $(x+2)y' + y = x^2 - 4$.

Note: $(x+2)y' + y = x^2 - 4 \quad \Rightarrow \quad y' + \frac{1}{(x+2)} \cdot y = \frac{1}{(x+2)} \cdot (x+2)(x-2)$

$$\Rightarrow \quad y' + \frac{1}{(x+2)} \cdot y = (x-2)$$

Note: The differential equation is of the form $\frac{dy}{dx} + M(x) \cdot y = N(x)$.

a) $e^{\int M(x) \cdot dx} = \underline{\quad ? \quad}$

b) $\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx = \underline{\quad ? \quad}$

c) $y = \dfrac{\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx \; + C}{e^{\int M(x) \cdot dx}} = \underline{\quad ? \quad}$

Question 6:

Solve the differential equation $(x-5)\,y' + y = x^2 - 25$.

Note: $(x-5)\,y' + y = x^2 - 25 \quad \Rightarrow \quad y' + \dfrac{1}{(x-5)} \cdot y = \dfrac{1}{(x-5)} \cdot (x+5)(x-5)$

$$\Rightarrow \quad y' + \frac{1}{(x-5)} \cdot y = x + 5$$

Note: The differential equation is of the form $\dfrac{dy}{dx} + M(x) \cdot y = N(x)$.

a) $e^{\int M(x)\cdot dx} = \underline{\quad ? \quad}$

b) $\int N(x) \cdot e^{\int M(x)\cdot dx} \cdot dx = \underline{\quad ? \quad}$

c) $y = \dfrac{\int N(x) \cdot e^{\int M(x)\cdot dx} \cdot dx \; + C}{e^{\int M(x)\cdot dx}} = \underline{\quad ? \quad}$

Question 7:

Solve the differential equation $(x-7)y'+y=\dfrac{x-7}{x+3}$.

Note: $(x-7)y'+y=\dfrac{x-7}{x+3} \Rightarrow \quad y'+\dfrac{1}{(x-7)}\cdot y=\dfrac{1}{(x-7)}\cdot\dfrac{x-7}{x+3} \Rightarrow \quad y'+\dfrac{1}{(x-7)}\cdot y=\dfrac{1}{x+3}$

Note: The differential equation is of the form $\dfrac{dy}{dx}+M(x)\cdot y=N(x)$.

$M(x)=\dfrac{1}{(x-7)}$ and $=\dfrac{1}{x+3}$.

a) $e^{\int M(x)\cdot dx} =$ ___?___

b) $\int N(x)\cdot e^{\int M(x)\cdot dx}\cdot dx =$ ___?___

c) $y=\dfrac{\int N(x)\cdot e^{\int M(x)\cdot dx}\cdot dx \;+C}{e^{\int M(x)\cdot dx}} =$ ___?___ .

Question 8:

Find the general solution of the differential equation $5y' + 2x^2 \cdot y = x^2 + 4$.

Multiply by $\frac{1}{5}$: $\frac{1}{5} \cdot 5y' + \frac{1}{5} \cdot 2x^2 \cdot y = \frac{1}{5}x^2 + \frac{1}{5} \cdot 4 \quad \Rightarrow \quad y' + \frac{2}{5} \cdot x^2 \cdot y = \frac{1}{5}x^2 + \frac{4}{5}$

Note: The differential equation $y' + \frac{2}{5} \cdot x^2 \cdot y = \frac{1}{5}x^2 + \frac{4}{5}$ is of the form $\frac{dy}{dx} + M(x) \cdot y = N(x)$.

$M(x) = \frac{2}{5} \cdot x^2$ and $N(x) = \frac{1}{5}x^2 + \frac{4}{5}$

a) $e^{\int M(x) \cdot dx} =$ ___?___

b) $\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx =$ ___?___

c) $y = \dfrac{\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx + C}{e^{\int M(x) \cdot dx}} =$ ___?___

d) Find the particular solution that satisfies intial condition $y(1) = 2$.

Note: When $x = 1$, $y = 2$

Question 9:

Find the general solution of the differential equation $4x^2y' + 2x \cdot y = x + 4$.

Multiply by $\frac{1}{4x^2}$: $\quad \frac{1}{4x^2} \cdot 4x^2y' + \frac{1}{4x^2} \cdot 2x \cdot y = \frac{1}{4x^2} \cdot x + \frac{1}{4x^2} \cdot 4 \quad \Rightarrow \quad y' + \frac{1}{2x} \cdot y = \frac{1}{4x} + \frac{1}{x^2}$

Note: The differential equation $y' + \frac{1}{2x} \cdot y = \frac{1}{4x} + \frac{1}{x^2}$ is of the form $\frac{dy}{dx} + M(x) \cdot y = N(x)$.

$M(x) = \frac{1}{2x}$ and $N(x) = \frac{1}{4x} + \frac{1}{x^2}$

a) $e^{\int M(x) \cdot dx} =$ ___?___

b) $\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx =$ ___?___

c) $y = \dfrac{\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx + C}{e^{\int M(x) \cdot dx}}$ = ___?___

d) Find the particular solution that satisfies intial condition $y(0) = 2$.

Note: When $x = 1$, $y = 2$

Question 10:

Find the general solution of the differential equation $y' + \frac{3}{x} \cdot y = \cos x$.

Note: The differential equation $y' + 2x^2 \cdot y = \cos x$ is of the form $\frac{dy}{dx} + M(x) \cdot y = N(x)$.

$M(x) = 2x^2$ and $N(x) = \cos x$

a) $e^{\int M(x) \cdot dx} = \underline{\quad ? \quad}$

b) $\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx = \underline{\quad ? \quad}$

c) $y = \dfrac{\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx + C}{e^{\int M(x) \cdot dx}} = \underline{\quad ? \quad}$

d) Find the particular solution that satisfies intial condition $y(\pi) = 2$.

Note: When $x = \pi$, $y = 2$

Question 11:

Find the general solution of the differential equation $y' + \frac{1}{x} \cdot y = 4\sin x$.

Note: $y' + \frac{1}{x} \cdot y = 4\sin x \quad \Rightarrow \quad \frac{dy}{dx} + \frac{1}{x} \cdot y = 4\sin x$

Note: The differential equation is of the form $\frac{dy}{dx} + M(x) \cdot y = N(x)$.

$M(x) = \frac{1}{x}$ and $N(x) = 4\sin x$

a) $e^{\int M(x) \cdot dx} = e^{\int \frac{1}{x} \cdot dx} =$ ____?____

b) $\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx =$ ____?____

c) $y = \dfrac{\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx \ + C}{e^{\int M(x) \cdot dx}} =$ ____?____

d) Find the particular solution that satisfies intial condition $y(\pi/2) = 1$.

Note: When $x = \pi/2$, $y = 1$

Question 12:

Find the general solution of the differential equation $y' + \frac{2}{x} \cdot y = \frac{1}{x^2(1+\tan x)}$

Note: The differential equation $y' + \frac{2}{x} \cdot y = \frac{1}{x^2(1+\tan x)}$ is of the form $\frac{dy}{dx} + M(x) \cdot y = N(x)$.

$M(x) = \frac{2}{x}$ and $N(x) = \frac{1}{x^2(1+\tan x)}$

a) $e^{\int M(x) \cdot dx} = \underline{\quad ? \quad}$

b) $\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx = \underline{\quad ? \quad}$

c) $y = \frac{\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx + C}{e^{\int M(x) \cdot dx}} = \underline{\quad ? \quad}$

d) Find the particular solution that satisfies intial condition $y(\pi) = 1$.

Note: When $x = \pi$, $y = 1$

Question 13:

Find the general solution of the differential equation $x^2 \cdot y' + 4 \cdot x \cdot y = \dfrac{\sin x}{x}$.

Multiply by $\dfrac{1}{x^2}$: $\dfrac{1}{x^2} \cdot x^2 \cdot y' + \dfrac{1}{x^2} \cdot 4 \cdot x \cdot y = \dfrac{1}{x^2} \cdot \dfrac{\sin x}{x} \quad \Rightarrow \quad y' + \dfrac{4}{x} \cdot y = \dfrac{\sin x}{x^3}$

Note: The differential equation $y' + \dfrac{4}{x} \cdot y = \dfrac{\sin x}{x^3}$ is of the form $\dfrac{dy}{dx} + M(x) \cdot y = N(x)$.

$M(x) = \dfrac{4}{x}$ and $N(x) = \dfrac{\sin x}{x^3}$

a) $e^{\int M(x) \cdot dx} = \underline{\quad ? \quad}$

b) $\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx = \underline{\quad ? \quad}$

c) $y = \dfrac{\int N(x) \cdot e^{\int M(x) \cdot dx} \cdot dx + C}{e^{\int M(x) \cdot dx}} = \underline{\quad ? \quad}$

d) Find the particular solution that satisfies intial condition $y(\pi) = 1$.

Note: When $x = \pi$, $y = 1$

Question 14:

Find the general solution of the differential equation $y' + (3\tan x)\cdot y = \cos x$.

Note: The differential equation $y' + (3\tan x)\cdot y = \cos x$. is of the form $\frac{dy}{dx} + M(x)\cdot y = N(x)$.

$M(x) = 3(\tan x)$ and $N(x) = \cos x$

a) $e^{\int M(x)\cdot dx} = \underline{\quad ? \quad}$

b) $\int N(x)\cdot e^{\int M(x)\cdot dx}\cdot dx = \underline{\quad ? \quad}$

c) $y = \dfrac{\int N(x)\cdot e^{\int M(x)\cdot dx}\cdot dx + C}{e^{\int M(x)\cdot dx}} = \underline{\quad ? \quad}$

d) Find the particular solution that satisfies intial condition $y(\pi) = 1$.

Note: When $x = \pi$, $y = 1$